

EDUCATION	Northeastern University	China	Computer Science	BS, 2003
	Katholieke Universiteit Leuven	Belgium	Statistics	MS, 2005
	Emory University	Atlanta	Computer Science	MS, 2008
	Illinois Institute of Technology	Chicago	Computer Science	PhD, 2015
	University of Washington	Seattle	Database Systems	Postdoc, 2016

EMPLOYMENT	IBM Almaden Research Center, San Jose, CA	Research Intern, 2015
	Argonne National Laboratory, Argonne, IL	Research Intern, 2014
	Pacific Northwest National Laboratory, Richland, WA	Research Intern, 2013
	Epic Systems, Madison, WI	Software Engineer, 2009-11
	IDMI Systems, Atlanta, GA	Software Engineer, 2008-09

PUBLICATIONS ——— Editorials ———

- [42] Kyle Chard, Bogdan Nicolae, Alexandru Costan, and **Dongfang Zhao**. Proceedings of the ACM 8th Workshop on Scientific Cloud Computing, ScienceCloud@HPDC 2017, Washington D.C., USA, June 2017.
- [41] Kyle Chard, Alexandru Costan, Bogdan Nicolae, and **Dongfang Zhao**. Proceedings of the ACM 7th Workshop on Scientific Cloud Computing, ScienceCloud@HPDC 2016, Kyoto, Japan, June 2016.

——— Book Chapters ———

- [40] **Dongfang Zhao**, Akash Mahakode, Sandip Lakshminarasiah, and Ioan Raicu. High-Performance Storage Support for Scientific Big Data Applications on the Cloud. *Resource Management for Big Data Platforms: Algorithms, Modelling, and High-Performance Computing Techniques*. Computer Communications and Networks (Springer), pp 147-170, October 2016.

——— Journals ———

- [39] **Dongfang Zhao**, Kan Qiao, Zhou Zhou, Tonglin Li, Zhihan Lu, and Xiaohua Xu. Toward Efficient and Flexible Metadata Indexing of Big Data Systems. *IEEE Transactions on Big Data (TBD)*. Accepted, December 2016.
- [38] **Dongfang Zhao**, Ke Wang, Kan Qiao, Tonglin Li, Iman Sadooghi, and Ioan Raicu. Toward High-performance Key-value Stores through GPU Encoding and Locality-aware Encoding. *Journal of Parallel and Distributed Computing (JPDC)*. 96(10):27-37, October 2016.
- [37] Zhou Zhou, Xu Yang, **Dongfang Zhao**, Paul Rich, Wei Tang, Jia Wang, Zhiling Lan. I/O-Aware Batching Scheduling and Bandwidth Allocation for Petascale Computing Systems. *Journal of Parallel Computing (PARCO)*. 58:107-116, October 2016.
- [36] **Dongfang Zhao**, Kan Qiao, Zhou Zhou, Tonglin Li, Xiaobing Zhou, and Ioan Raicu. Exploiting Multi-Cores for Efficient Interchange of Large Messages in Distributed Systems. *Concurrency and Computation: Practice and Experience (CCPE)*. 28(13):3568–3585, September 2016.
- [35] **Dongfang Zhao**, Ning Liu, Dries Kimpe, Robert Ross, Xian-He Sun, and Ioan Raicu. Towards Exploring Data-Intensive Scientific Applications at Extreme Scales through Systems and Simulations. *IEEE Transactions on Parallel and Distributed Systems (TPDS)*. 27(6):1824-1837, June 2016.

- [34] **Dongfang Zhao**, Kan Qiao, and Ioan Raicu. Towards Cost-Effective and High-Performance Caching Middleware for Distributed Systems. *International Journal of Big Data Intelligence (IJBDI)*. 3(2):92-110, June 2016.
 - [33] **Dongfang Zhao**, Kan Qiao, Jian Yin, and Ioan Raicu. Dynamic Virtual Chunks: On Supporting Efficient Accesses to Compressed Scientific Data. *IEEE Transactions on Services Computing (TSC)*. 9(1):96-109, February 2016.
 - [32] Tonglin Li, Xiaobing Zhou, Ke Wang, **Dongfang Zhao**, Iman Sadooghi, Zhao Zhang, and Ioan Raicu. A Convergence of Key-Value Storage Systems from Clouds to Supercomputers. *Concurrency and Computation: Practice and Experience (CCPE)*. 28(1):44-69, January 2016.
 - [31] **Dongfang Zhao** and Li Yang. Incremental Isometric Embedding of High-Dimensional Data Using Connected Neighborhood Graphs. *IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI)*. 31(1):86-98, January 2009.
- Conferences ——
- [30] Iman Sadooghi, Geet Kumar, Ke Wang, **Dongfang Zhao**, Tonglin Li, and Ioan Raicu. Albatross: an Efficient Cloud-enabled Task Scheduling and Execution Framework using Distributed Message Queues. *Proceedings of the 12th IEEE International Conference on eScience*. Baltimore, MD, USA, October 2016.
 - [29] **Dongfang Zhao**. Toward Real-time and Fine-grained Monitoring of Software-defined Networking in the Cloud. *Proceedings of the 9th IEEE International Conference on Cloud Computing (CLOUD)*. San Francisco, CA, USA, June 2016.
 - [28] Iman Sadooghi, Ke Wang, Shiva Srivastava, Dharmit Patel, **Dongfang Zhao**, Tonglin Li, and Ioan Raicu. FaBRiQ: Leveraging Distributed Hash Tables towards Distributed Publish-Subscribe Message Queues. *IEEE/ACM International Symposium on Big Data Computing*. Limassol, Cyprus, December 2015.
 - [27] **Dongfang Zhao**, Nagapramod Mandagere, Gabriel Alatorre, Mohamed Mohamed, and Heiko Ludwig. Toward Locality-aware Scheduling for Containerized Cloud Services. *Proceedings of the 2015 IEEE International Conference on Big Data*. Santa Clara, CA, USA, October 2015.
 - [26] Tonglin Li, Ke Wang, Shiva Srivastava, **Dongfang Zhao**, Kan Qiao, Iman Sadooghi, Xiaobing Zhou, and Ioan Raicu. A Flexible QoS Fortified Distributed Key-Value Storage System for the Cloud. *Proceedings of the 2015 IEEE International Conference on Big Data*. Santa Clara, CA, USA, October 2015.
 - [25] Zhou Zhou, Xu Yang, **Dongfang Zhao**, Paul Rich, Wei Tang, Jia Wang, and Zhiling Lan. I/O-Aware Batch Scheduling for Petascale Computing Systems. *Proceedings of the 2015 IEEE International Conference on Cluster Computing*. Chicago, IL, USA, September 2015.
 - [24] **Dongfang Zhao**, Jian Yin, Kan Qiao, and Ioan Raicu. Virtual Chunks: On Supporting Random Accesses to Scientific Data in Compressible Storage Systems. *Proceedings of the 2014 IEEE International Conference on Big Data*. Washington, DC, USA, October 2014.
 - [23] Ke Wang, Xiaobing Zhou, Tonglin Li, **Dongfang Zhao**, Michael Lang, and Ioan Raicu. Optimizing Load Balancing and Data-Locality with Data-Aware Scheduling. *Proceedings of the 2014 IEEE International Conference on Big Data*. Washington, DC, USA, October 2014.

- [22] **Dongfang Zhao**, Zhao Zhang, Xiaobing Zhou, Tonglin Li, Ke Wang, Dries Kimpe, Philip Carns, Robert Ross, and Ioan Raicu. FusionFS: Towards Supporting Data-Intensive Scientific Applications on Extreme-Scale High-Performance Computing Systems. *Proceedings of the 2014 IEEE International Conference on Big Data*. Washington, DC, USA, October 2014.
 - [21] **Dongfang Zhao**, Kan Qiao, and Ioan Raicu. HyCache+: Towards Scalable High-Performance Caching Middleware for Parallel File Systems. *Proceedings of the 14th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGrid)*. Chicago, IL, USA, May 2014.
 - [20] **Dongfang Zhao**, Chen Shou, Tanu Malik, and Ioan Raicu. Distributed Data Provenance for Large-Scale Data-Intensive Computing. *Proceedings of the 2013 IEEE International Conference on Cluster Computing*. Indianapolis, IN, USA, September 2013.
 - [19] **Dongfang Zhao**, Kent Burlingame, Corentin Debains, Pedro Alvarez-Tabio, and Ioan Raicu. Towards High-Performance and Cost-Effective Distributed Storage Systems with Information Dispersal Algorithms. *Proceedings of the 2013 IEEE International Conference on Cluster Computing*. Indianapolis, IN, USA, September 2013.
 - [18] Tonglin Li, Xiaobing Zhou, Kevin Brandstatter, **Dongfang Zhao**, Ke Wang, Anupam Rajendran, Zhao Zhang, and Ioan Raicu. ZHT: A Light-Weight Reliable Persistent Dynamic Scalable Zero-Hop Distributed Hash Table. *Proceedings of the 27th IEEE International Parallel & Distributed Processing Symposium (IPDPS)*. Boston, MA, USA, May 2013.
 - [17] **Dongfang Zhao**, Da Zhang, Ke Wang, and Ioan Raicu. Exploring Reliability of Exascale Systems Through Simulations. *Proceedings of the 21st ACM/SCS High Performance Computing Symposium (HPC)*. San Diego, CA, USA, April 2013.
 - [16] Robin Lohfert, James Lu, and **Dongfang Zhao**. Solving SQL Constraints by Incremental Translation to SAT. *Proceedings of the 21st International Conference on Industrial, Engineering and Other Applications of Applied Intelligent Systems (IEA/AIE)*. Wroclaw, Poland, June 2008.
 - [15] **Dongfang Zhao** and Li Yang. Incremental Construction of Neighborhood Graphs for Nonlinear Dimensionality Reduction, *Proceedings of the 18th International Conference on Pattern Recognition (ICPR)*. Hong Kong, China, August 2006.
- Workshops ——
- [14] **Dongfang Zhao**, Xu Yang, Iman Sadooghi, Gabriele Garzoglio, Steven Timm, and Ioan Raicu. High-Performance Storage Support for Scientific Applications on the Cloud. *ScienceCloud workshop, proceedings of the 24th ACM International Symposium on High Performance Distributed Computing (HPDC)*. Portland, OR, USA, June 2015.
 - [13] Tonglin Li, Kate Keahey, Ke Wang, **Dongfang Zhao**, and Ioan Raicu. A Dynamically Scalable Cloud Data Infrastructure for Sensor Networks. *ScienceCloud workshop, proceedings of the 24th ACM International Symposium on High Performance Distributed Computing (HPDC)*. Portland, OR, USA, June 2015.
 - [12] **Dongfang Zhao** and Ioan Raicu. Storage Support for Data-Intensive Applications on Extreme-Scale HPC Systems. *Doctoral Showcase, the 2014 ACM/IEEE Conference on Supercomputing (SC)*. New Orleans, LA, USA, November 2014.

- [11] **Dongfang Zhao** and Ioan Raicu. HyCache: A User-Level Caching Middleware for Distributed File Systems. *Proceedings of International Workshop on High Performance Data Intensive Computing, collocated with IEEE International Symposium on Parallel and Distributed Processing (IPDPS)*. Boston, MA, USA, May 2013.
- [10] Chen Shou, **Dongfang Zhao**, Tanu Malik, and Ioan Raicu. Towards a Provenance-Aware Distributed Filesystem. *The 5th Workshop on the Theory and Practice of Provenance, collocated with USENIX Symposium on Networked Systems Design and Implementation (NSDI)*. Lombard, IL, USA, April 2013.
- [9] **Dongfang Zhao** and Ioan Raicu. Distributed File Systems for Exascale Computing. *Doctoral Showcase, the 2012 ACM/IEEE Conference on Supercomputing (SC)*. Salt Lake City, UT, USA, November 2012.

—— Extended Abstracts ——

- [8] Parmita Mehta, Sven Dorkenwald, **Dongfang Zhao**, Magdalena Balazinska, Alvin Cheung, and Ariel Rokem. Comparing Systems for Analyzing Big Neuroscience Imaging Data. *NeuroFutures*. Seattle, WA, USA, June 2016.
- [7] Jian Yin and **Dongfang Zhao**. Data Confidentiality Challenges in Big Data Applications. *Proceedings of the 2015 IEEE International Conference on Big Data*. Santa Clara, CA, USA, October 2015.
- [6] Xiaobing Zhou, Tonglin Li, Ke Wang, **Dongfang Zhao**, Iman Sadooghi, and Ioan Raicu. MHT: A Light-weight Scalable Zero-hop MPI Enabling Distributed Key-Value Store. *Proceedings of the 2015 IEEE International Conference on Big Data*. Santa Clara, CA, USA, October 2015.
- [5] Tonglin Li, Chaoqi Ma, Jiabao Li, Xiaobing Zhou, Ke Wang, **Dongfang Zhao**, and Ioan Raicu. GRAPH/Z: A Key-Value Store Based Scalable Graph Processing System. *Proceedings of the 2015 IEEE International Conference on Cluster Computing*. Chicago, IL, USA, September 2015.
- [4] **Dongfang Zhao** and Ioan Raicu. Storage Support for Data-Intensive Scientific Applications on the Cloud. *NSFCloud Workshop on Experimental Support for Cloud Computing*. Arlington, VA, USA, December 2014.
- [3] **Dongfang Zhao**, Jian Yin, and Ioan Raicu. Improving the I/O Throughput for Data-Intensive Scientific Applications with Efficient Compression Mechanisms. *The 2013 ACM/IEEE Conference on Supercomputing (SC)*. Denver, CO, USA, November 2013.

—— Theses ——

- [2] **Dongfang Zhao**. Big Data System Infrastructure at Extreme Scales. *Doctoral Dissertation of Computer Science*. Department of Computer Science, Illinois Institute of Technology, USA, July 2015.
- [1] **Dongfang Zhao**. Adaptive Weights Smoothing and Applications. *Master's Thesis of Statistics*. Department of Statistics, Katholieke Universiteit Leuven, Belgium, July 2005.

SERVICES

Journal Editorial Board

Journal of Big Data (Springer)	2015-17
Big Data and Cognitive Computing (MPDI)	2016-17

Conference Co-Chairs

ACM Workshop on Scientific Cloud Computing	2016-17
IEEE International Conference on Networking, Architecture, and Storage	2016
ACM/IEEE International Symposium on Big Data Computing	2015

Journal Referee (selected)

IEEE Transactions on Parallel and Distributed Systems	2014-16
IEEE Transactions on Cloud Computing	2014-16
Journal of Big Data (Springer)	2015-16
Journal of Cluster Computing (Springer)	2015-16
Journal of Grid Computing (Springer)	2015-16
Journal of Parallel Computing (Elsevier)	2015-16
Future Generation Computer Systems (Elsevier)	2015-16
Concurrency and Computation: Practice and Experience (Wiley)	2015-17

Conference Reviewer / Technical Program Committee (selected)

ACM/IEEE Supercomputing Conference (SC)	2015-16
IEEE Int'l Parallel & Distributed Processing Symposium (IPDPS)	2014-16
IEEE Symposium on Cluster, Cloud and Grid Computing (CCGrid)	2013-16
IEEE Int'l Conf. on Cloud Computing Technology & Science (CloudCom)	2014-15
ACM Symposium on High-Performance Distributed Computing (HPDC)	2012-14

University and Department Services

Committee of Undergraduate, CSE Department, Nevada	2016-17
Committee of Web, CSE Department, Nevada	2016-17

TEACHING

Teaching Assistant / Instructor

CS491/691 Data Intensive Computing, Nevada	Fall 2017
CS554 Data Intensive Computing, IIT	Fall 2013, Spring 2015
CS553 Cloud Computing, IIT	Spring 2013, Fall 2014
CS495 Introduction to Distributed Systems, IIT	Fall 2012

Student Supervision (Name, Program, School, Employment, Semester)

Kevin Brandstatter, BS/MS Computer Science, IIT, Epic	Spring 2015
Daniel Gordon, BS Computer Science, IIT, Nokia	Fall 2014
Chen Shou, MS Computer Science, IIT, Amazon	Fall 2013
Kent Burlingame, BS Computer Science, IIT, Microsoft	Spring 2013
Da Zhang, MS Computer Science, IIT, Virginia Tech PhD program	Fall 2012

REFERENCES

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 Computer Science & Engineering Department, University of Washington, Seattle, WA

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 Department of Computer Science, Illinois Institute of Technology, Chicago, IL

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